Pincushion And Barrel Distortion

This is a problem with almost every CRT monitor. If you thought the factory default settings get you straight lines, you will be disappointed to find that these settings are hardly optimised.

Pincushion is a form of distortion where lines bend inwards. The opposite of this is barrel distortion. There are multiple reasons why these effects happen, the primary one being that the screen is constructed almost flat, disturbing its natural spherical shape. With this in mind, monitors come with special circuitry that corrects pincushion and barrel distortion. However, not all the monitors displayed straight lines even after trying every trick in the book to calibrate them.

In the 17-inch category, the NEC MultiSync FE791SB, Samsung 793S and LG Flatron E700SH were the ones that returned the best results. The results, after all the adjustments, were not perfect, but they were acceptable to the point where only graphics professionals could make out flaws. Users working with office applications and surfing the Net would find the results impressive.

The HCL785RFM and the Philips 107 E5 failed to display horizontal lines straight. The HCL and the Philips monitors failed to provide the necessary tools for perfect adjustment, and scored lower on this test. If your work involves working with graphics or industrial design, we recommend the NEC MultiSync FE791SB.

How We Tested

Our test bed consisted of a Pentium 4 2.8 GHz equipped with an nVidia GeForceFX 5950 Ultra graphics card. The system ran Windows XP SP2. The CRT monitors were tested at their native resolutions (1024 x 768 at 75 Hz for 17-inch, and 1280 x 1024 at 75 Hz for 19-inch monitors).

We adjusted each display to perform optimally in the viewing conditions, using manufacturer-recommended or Windows' default settings when possible. We also had a reference monitor, which was evaluated first, and then the rest of the monitors were compared to the image displayed by the reference monitor.

DisplayMate Tests

With help from DisplayMate Technologies, we ran through our own scripted selection of test screens in DisplayMate Multimedia Edition. These screens help us isolate common phenomena such as flicker, moiré, bounce, ghosting, and convergence errors. We broke down our own analysis into five categories: focus, regulation, geometry, colour quality and uniformity, reverse text, moiré and interference. A product's scores in each category was compiled and averaged to produce a single performance rating. The ratings you see in our reviews are a composite of the battery of individual screen analyses we ran for each monitor.

Our test script, described below, represents only our base-testing. Many of the screens in this test suite can be configured in a variety of ways, including with different background and foreground colours. Depending on the characteristics of an individual display, we used several variations of the screens listed below, along with other DisplayMate screens.

Resolution Tests

Horizontal Line Resolution: This screen contains five horizontal bars composed of vertical white lines in varying frequencies of lines per inch. The higher line frequencies demand more from the display's horizontal resolution capabilities. We used this screen to adjust the sharpness of the monitor and evaluate its horizontal resolution capabilities.

Vertical Line Resolution: This screen is similar to the previous test, and contains five horizontal bars composed of horizontal white lines in varying frequencies of lines per inch. The higher number of lines per inch areas stressed the display's vertical resolution capabilities.

Focus Matrix: The graphic pattern on this screen reveals variations in focus across the display. This screen was used to adjust and to test image focus. It is also handy for checking geometric linearity, especially in the corners.

Page Of Text: This screen allows us to examine the display's ability to render text under a variety of conditions. With it, you can cycle through various text and background colours, view split screens with inverse text and background colours, and alter the type and size of a font. We found it especially useful in evaluating a display's ability to render anti-aliasing enhancements.

Sharpness And Resolution Tests

Midrange Streaking: As the title suggests, this screen allows us to check the display's propensity for streaking and ghosting, light or dark shadows that trail an image in areas where large changes in intensity are present. You may detect this on your own monitor when it renders large, chunky graphics elements such as bar graphs or tiled arrangements of open windows.

Fine Focus And Resolution Matrix: Moiré looks like a tiger's stripes, or like ripples, superimposed on a CRT monitor's screen. Though it's found on virtually all CRTs, it's actually less common in consumer CRTs because they are generally not as sharp as higher-end displays. Some monitors include moiré-reduction features.

Geometry Tests

Pincushion And Barrel Distortion: This screen is used to measure the geometric distortion displayed by the CRT monitor after calibrating it using its OSD menu.

Screen Regulation: This screen helps detect image contraction and expansion with strong changes in brightness. It displays a large, flashing square against a black background. The square can be assigned various colours, but white is generally the most useful. On a poor-quality CRT, the square appears to grow slightly as it flashes.

Colour And Greyscale Tests

Streaking And Ghosting: In this test, various screens comprising grey and colour bars are displayed against a grey background. Streaking and ghosting usually occur when there is a change in the intensity between adjacent areas. We looked for shadows trailing the edges of bars and blocks.

Level Shift: This, again, occurs at the edge of areas with varying brightness. In white and black level shift, the area adjacent to the dominant white or black area will have a blurry edge—displaying a red, green or blue edge.

Miscellaneous Tests

Background Noise And Interference (Screen 10:3): We examined this screen for a list of things that should not be seen. These may include irregularities relating to noise aggravated by poor-quality cables or other problems as are normally associated with the electronics of the display.

Defocusing Blooming And Halos: This slide displays lines and rings of varying brightness. The monitor should be able to satisfactorily produce the image, and the bright part should not look washed out or appear thick.

Features Test: In the features test, we looked for the set of features the monitor provided—such as a flat screen, video bandwidth, the maximum horizontal and vertical frequency supported by the particular monitor, dimensions, weight, and other relevant aspects. The monitors were awarded points according to the extra features they offered.